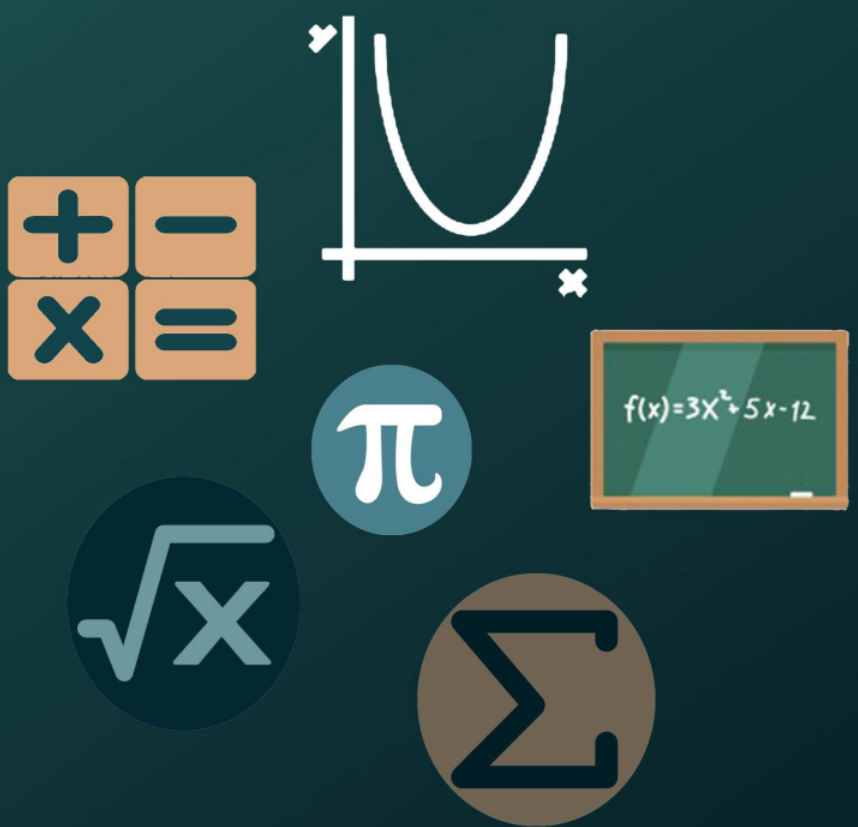
MATHEMATICS

Differential Equation



SHEET

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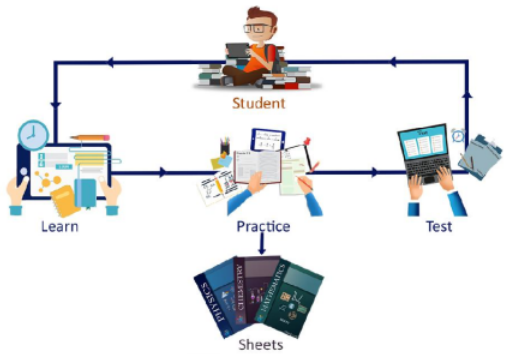
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Order & Degree of Differential Equation	1-4	1-2, 6-8, 16-19	1	
2	Formation of Differential Equation		20,21	2-4	
3	Solution of Differential Equation	5	22-25	5,6	1
4	Variable Seperable Form Differential Equation	6-8	26-29	7-10	22
5	Equations Reducible to Variable Seperable Form	9-11	30,31		
6	Homogeneous Form of Differential Equation	12, 13	3	11-14	2,15
7	Equations Reducible to Homogeneous Form		4	15	
8	Linear Differential Equation	14-20	32,33		
9	Differential Equations Reducible to linear Form		5, 34	16,17	3
10	Exact Differential Equation	21,22	9-11		16
11	Miscellaneous Problems of Differential Equations		12-14	18,19	4-16, 17-21, 23-30
12	Higher Order & Higher Degree Differential Equations		35		
13	Use of Polar Coordinates to Solve D.E.'s		15		

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Order & Degree of Differential Equation

Q1. The degree of the differential equation $y_2^{3/2} - y_1^{1/2} - 4 = 0$ is-

(A) 6 (B) 3
(C) 2 (D) 4

Q2. The differential equation whose solution is y = Asin x + B cos x, is

(A) $\frac{d^2y}{dx^2} + y = 0$ (B) $\frac{d^2y}{dx^2} - y = 0$

(C) $\frac{dy}{dx} + y = 0$ (D) None of these

Q3. The differential equation corresponding to the curves y = a cos μx + b sin μx, where a, b are arbitrary constants, is-

(A) $y_2 = y_1$ (B) $y_2 = -y$
(C) $y_2 = -\mu y$ (D) $y_2 + \mu^2 y = 0$

Q4. The differential equation of all the lines in the xy-plane is

(A) $\frac{dy}{dx} - x = 0$ (B) $\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$

(C) $\frac{d^2y}{dx^2} = 0$ (D) $\frac{d^2y}{dx^2} + x = 0$

Solution of Differential Equation

Q5. The solution of the differential equation $\frac{dy}{dx} = e^x + \cos x + x + \tan x$ is-

(A) $y = e^x + \sin x + \frac{x^2}{2} + \log \cos x + c$

(B) $y = e^x + \sin x + \frac{x^2}{2} + \log \sec x + c$

(C) $y = e^x - \sin x + \frac{x^2}{2} + \log \cos x + c$

(D) $y = e^x - \sin x + \frac{x^2}{2} + \log \sec x + c$

Variable Seperable Differential Equation

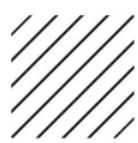
Q6. The general solution of differential equation $\frac{dy}{dx} = \log x$ is-

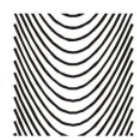
(A) y = x (log x + 1) + C
(B) y + x (logx+1) = C
(C) y = x (log x –1) + C
(D) None of these

Q7. If $\frac{dy}{dx} + \frac{1}{\sqrt{1-x^2}} = 0$, then-

(A) $y + \sin^{-1} x = C$
(B) $y^2 + 2\sin^{-1}x + c = 0$
(C) $x + \sin^{-1} y = 0$
(D) $x^2 + 2\sin^{-1} y = 1$

Q8. The general solution of the differential equation $\frac{dy}{dx} = \frac{1-x}{y}$ is a family of curves which looks most like which of the following ?

(A)  (B) 

(C)  (D) 

Equations Reducible to Variable Seperable Form

Q9. The solution of

$\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ is -

(A) $\log\left[1+\tan\left(\frac{x+y}{2}\right)\right]+c=0$

(B) $\log\left[1+\tan\left(\frac{x+y}{2}\right)\right]=x+c$

(C) $\log\left[1-\tan\left(\frac{x+y}{2}\right)\right]=x+c$

(D) None of these

Q10. The solution of the differential equation

$(x^2 - yx^2)\frac{dy}{dx} + y^2 + xy^2 = 0$ is -

(A) $\log\left(\frac{x}{y}\right)=\frac{1}{x}+\frac{1}{y}+c$

(B) $\log\left(\frac{y}{x}\right)=\frac{1}{x}+\frac{1}{y}+c$

(C) $\log(xy)=\frac{1}{x}+\frac{1}{y}+c$

(D) $\log(xy)+\frac{1}{x}+\frac{1}{y}=c$

Q11. The solution of the differential equation

$\frac{dy}{dx}=\frac{(1+x)y}{(y-1)x}$ is -

(A) $\log xy + x + y + c$

(B) $\log\left(\frac{x}{y}\right)+x-y=c$

(C) $\log xy + x - y = c$

(D) None of these

Homogeneous Form of Differential Equation

Q12. Integral curve satisfying $y' = \frac{x^2+y^2}{x^2-y^2}$, $y(1) = 2$ has the slope at the point (1, 0) of the curve equal to-

(A) $-\frac{5}{3}$ (B) -1

(C) 1 (D) $\frac{5}{3}$

Q13. The slope of the tangent at (x, y) to a curve passing through a point (2, 1) is $\frac{x^2+y^2}{2xy}$, then the equation of the curve is -

(A) $2(x^2 - y^2) = 3x$

(B) $2(x^2 - y^2) = 6y$

(C) $x(x^2 - y^2) = 6$

(D) $x(x^2 + y^2) = 10$

Linear Differential Equation

Q14. Solution of the differential equation, $y\,dx - x\,dy + xy^2dx = 0$ can be-

(A) $2x + x^2y = \lambda y$

(B) $2y + y^2x = \lambda y$

(C) $2y - y^2x = \lambda y$

(D) None of these

Q15. The solution of the equation $x\frac{dy}{dx} + 3y = x$ is-

(A) $x^3y+\frac{x^4}{4}+c=0$

(B) $x^3y-\frac{x^4}{4}=c$

(C) $x^3y+\frac{x^4}{4}=0$

(D) None of these

Q16. The solution of the differential equation $x\frac{dy}{dx}+y=x^2+3x+2$ is-

(A) $xy=\frac{x^3}{3}+\frac{3}{2}x^2+2x+c$

(B) $xy=\frac{x^4}{4}+x^3+x^2+c$

(C) $xy=\frac{x^4}{4}+\frac{x^3}{3}+x^2+c$

(D) $xy=\frac{x^4}{4}+x^3+x^2+cx$

Q17. Which of the following equation is linear-

(A) $\left(\frac{d^2y}{dx^2}\right)^2+x^2\left(\frac{dy}{dx}\right)^2=0$

(B) $y=\frac{dy}{dx}+\sqrt{1+\left(\frac{dy}{dx}\right)^2}$

(C) $\frac{dy}{dx}+\frac{y}{x}=\log x$

(D) $y\frac{dy}{dx}-4=x$

Q18. A solution of differential equation

$(\sec^2 y)\frac{dy}{dx}+2x\tan y=x^3$ is -

(A) $2\tan y=c\cdot e^{-x^2}+x^2-1$

(B) $\tan y=ce^{-x^2}+x^2-1$

(C) $\tan y=ce^{x^2}+x^2-1$

(D) None of these

Q19. A function $y=f(x)$ satisfies the condition $f'(x)\sin x+f(x)\cos x=1, f(x)$ being bounded when $x\to 0$.If $I=\int_0^{\pi/2} f(x)\,dx$ then-

(A) $\frac{\pi}{2}<I<\frac{\pi^2}{4}$ (B) $\frac{\pi}{4}<I<\frac{\pi^2}{2}$

(C) $1<I<\frac{\pi}{2}$ (D) $0<I<1$

Q20. If the differentiable equation

$\frac{dy}{dx}-y=y^2(\sin x+\cos x)$ with $y(0)=1$ then $y(\pi)$ has the value equal to

(A) e^{π} (B) $-e^{\pi}$

(C) $e^{-\pi}$ (D) $-e^{-\pi}$

Exact Differential Equation

Q21. The solution of the differential equation, $x^2\frac{dy}{dx}.\cos\frac{1}{x}-y\sin\frac{1}{x}=-1$, where $y\to -1$ as $x\to\infty$ is

(A) $y=\sin\frac{1}{x}-\cos\frac{1}{x}$

(B) $y=\frac{x+1}{x\sin\frac{1}{x}}$

(C) $y=\cos\frac{1}{x}+\sin\frac{1}{x}$

(D) $y=\frac{x+1}{x\cos\frac{1}{x}}$

Q22. In a chemical reaction a substance changes into another such that the rate of decomposition of a chemical substance x present at instant t is proportional to x itself i.e. amount of unchanged substance still present. If half of the substance present initially has been converted at the end of 1 minute then the time t in minutes at the end of which 99 % of the substance will have changed lies in the interval.

(A) 5 and 6 (B) 6 and 7

(C) 7 and 8 (D) more than 10

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is **TRUE** ?

(A) $\frac{dy}{dx}+\sin\left(\frac{dy}{dx}\right)=0$;

order = 1, degree = Not defined

(B) $\frac{d^5y}{dx^5}+e^{\left(\frac{dy}{dx}\right)}+y^2=0$;

order = 5 ; degree = 1

(C) $y = Px + \sqrt{a^2P^2+b^2}$;

where $P=\frac{dy}{dx}$ order = 1 ; degree = 2

(D) $\frac{d^2y}{dx^2}=\left\{1+\left(\frac{dy}{dx}\right)^4\right\}^{5/3}$;

order = 2; degree = 3

Q2. If order and degree of the differential equation $\sqrt{\frac{d^2y}{dx^2}}=\sqrt[3]{\frac{dy}{dx}+3}$ be α & β respectively then $(\alpha + \beta)$ is -

(A) Prime Number

(B) Composite Number

(C) Even Number

(D) Odd number

Q3. Identify the statement(s) which is/are True.

(A) $f(x, y) = e^{y/x} + \tan\frac{y}{x}$ is a homogeneous function of degree zero

(B) $x \cdot \ln\frac{y}{x}\,dx+\frac{y^2}{x}\sin^{-1}\frac{y}{x}\,dy=0$ is a homogeneous differential equation of degree one

(C) $f(x, y) = x^2 + \sin x \cdot \cos y$ is not homogeneous function.

(D) $(x^2 + y^2)\,dx - (xy^2 - y^3)\,dy = 0$ is a homegeneous differential equation.

Q4. Solutions of the differential equation $x^2\left(\frac{dy}{dx}\right)^2+xy\left(\frac{dy}{dx}\right)-6y^2=0$ -

(A) $y = cx^2$ (B) $x^3y = c$

(C) $xy^3 = c$ (D) $y = cx$

Q5. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx}\cdot\sin x - y\cos x+\frac{\sin^2 x}{x^2}=0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is

(A) $\lim\limits_{x\to 0} f(x) = 1$

(B) $\int_0^{\pi/2} f(x)\,dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)\,dx$ is greater than unity

(D) $f(x)$ is an odd function

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 6 to 8)

The **Order** of the differential equation is the order of the highest derivative appearing in the equation and the **Degree** of a differential equation which can be written as polynomial in the derivatives in the degree of the derivative of the highest order occuring in it, after it has been expressed in a form free from radicals and fraction and if differential equation can not be written as a polynomial in the derivatives, then degree deos not defined but order defined.

Q6. The degree of the differential equation

$$x = 1 + \left(xy\frac{dy}{dx}\right) + \frac{x^2y^2}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!}\left(\frac{dy}{dx}\right)^3 + ...$$

is-

(A) 0 (B) 1

(C) 2 (D) not defined

Q7. The degree of the differential equation

$$\left(\frac{d^3y}{dx^3}\right)^{2/3} + 3\frac{d^2y}{dx^2} + \frac{dy}{dx} + 5 = 0$$ is-

(A) 1 (B) 2

(C) 3 (D) None of these

Q8. The order and degree of the differential equation whose solution is $y = cx + c^3 - 5c^{3/2} + 3$ where c is arbitrary constants, are-

(A) 1 and 3 (B) 1 and 4

(C) 1 and 5 (D) 1 and 6

Paragraph # 2 (Ques 9 to 11)

A curve $y = f(x)$ satisfies the differential equation $(1 + x^2)\frac{dy}{dx} + 2yx = 4x^2$ and passes through the origin.

Q9. The function $y = f(x)$

(A) is strictly increasing $\forall\ x \in R$.

(B) is such that it has a minima but no maxima.

(C) is such that it has a maxima but no minima.

(D) has no inflection point.

Q10. The area enclosed by $y = f^{-1}(x)$, the x-axis and the ordinate at $x = 2/3$ is

(A) $2\ ln\ 2$ (B) $\frac{4}{3}\ ln\ 2$

(C) $\frac{2}{3}\ ln\ 2$ (D) $\frac{1}{3}\ ln\ 2$

Q11. For the function $y = f(x)$ which one of the following does not hold good?

(A) $f(x)$ is a rational function

(B) $f(x)$ has the same domain and same range.

(C) $f(x)$ is a transcidental function

(D) $y = f(x)$ is a bijective mapping.

Paragraph # 3 (Ques 12 to 14)

A drop of liquid evaporates at a rate proportional to its surface area, if the radius initially is 5mm and 5 minute later the radius is reduced to 2mm.

On the basis of above information, answer the following questions-

Q12. The rate with which the surface area changes after 5 minutes is-

(A) $8\ \pi$ sq. mm/min

(B) $4.8\ \pi$ sq. mm/min

(C) $9.6\ \pi$ sq. mm/min

(D) $16\ \pi$ sq. mm/min

Q13. The rate of change of radius is –

(A) linear function of time

(B) circular function of time

(C) parabolic function of time

(D) independent of time

Q14. The rate of evaporation after, 5 minutes is

(A) $8\ \pi$ (B) $4.8\ \pi$

(C) $9.6\ \pi$ (D) $16\ \pi$

MATCH THE COLUMN TYPE

Q15. **Column I** **Column II**

(A) If the function $y = e^{4x} + 2e^{-x}$ is a solution of the differential equation $\frac{\frac{d^3y}{dx^3} - 13\frac{dy}{dx}}{y} = K$ then the value of $K/3$ is — (P) 3

(B) Number of straight lines which satisfy the differential equation $\frac{dy}{dx} + x\left(\frac{dy}{dx}\right)^2 - y = 0$ is — (Q) 4

(C) If real value of m for which the subtitution, $y = u^m$ will transform the differential equation, $2x^4y\frac{dy}{dx} + y^4 = 4x^6$ into a homogenous equation then the value of $2m$ is — (R) 2

(D) If the solution of differential equation $x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 12y$ is $y = Ax^m + Bx^{-n}$ then $|m + n|$ is — (S) 1

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column-I (Differential Equation)	**Column-II** (Order)	**Column-III** (Degree)
(A) $\{(y')^3 + y''\}^2 = a\,y'$	(P) 1	(L) 1
(B) $3y = 7x\,y' + \frac{5}{y'}$	(Q) 2	(M) 2
(C) $y'' - 5y' + 6y = 0$	(R) 3	(N) 3
(D) $(y' - 4x)^{3/2} = x + 5y'$	(S) 4	(O) not defined

Q16. Which of the following is only correct combination?

(A) A-P-M (B) A-Q-N

(C) A-Q-M (D) A-R-M

Q17. Which of the following is only correct combination?

(A) B-P-M (B) B-Q-N

(C) B-Q-M (D) B-R-M

Q18. Which of the following is only correct combination?

(A) C-P-O (B)C-Q-L

(C) C-Q-O (D) C-R-L

Q19. Which of the following is only correct combination?

(A) D-P-N (B) D-Q-L

(C) D-Q-M (D) D-R-L

SUBJECTIVE TYPE

Q20. Form the differential equation corresponding to $y^2 = n\,(b^2 - x^2)$ by eliminating n and b.

Q21. The differential equation of all circles which passes through the origin and whose centre lies on y-axis, is

Q22. Find the general solution of $x^2\dfrac{dy}{dx} = 6$

Q23. Verify that $y = cx + \dfrac{a}{c}$ is solution of the differential equation $y = x\dfrac{dy}{dx} + a\dfrac{dx}{dy}$

Q24. Find the solution of the differential equation

$$\frac{dy}{dx} = \operatorname{cosec} x\left[\cot x - \operatorname{cosec} x\right]$$

Q25. Find the solution of the differential equation

$$\frac{xdy}{x^2+y^2} = \left(\frac{y}{x^2+y^2} - 1\right)dx$$

Q26. Solve the differential equation

$$(x^3 - y^2x^3)\frac{dy}{dx} + y^3 + x^2y^3 = 0$$

Q27. Solve the differential equation

$$\frac{dy}{dx} = e^{x+y} + x^2\,e^y$$

Q28. Solve the differential equation

$$\frac{dy}{dx} = y^2\left(e^{2x} + 1\right)$$

Q29. Solve the differential equation

$$\frac{dy}{dx} + \frac{1+\cos 2y}{1-\cos 2x} = 0$$

Q30. Find the solution of differential equation

$$\frac{dy}{dx} = \cos(x + y)$$

Q31. Find the solution of differential equation

$(x + y)(dx - dy) = dx + dy$

Q32. Find the solution of differential equation

$$\cos x\frac{dy}{dx} + y\sin x = 1$$

Q33. Solve the differential equation

$$(1-x^2)\frac{dy}{dx} + 2xy = x\sqrt{1-x^2}$$

Q34. Solve the differential equation

$$x\frac{dy}{dx} + y = x^3y^6$$

Q35. Find the solution of the differential equation

$$\cos^2 x\,\frac{d^2y}{dx^2} = 1$$

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The order and degree of the differential equation $\sqrt[3]{\frac{dy}{dx}} - 4\frac{d^2y}{dx^2} - 7x = 0$ are a and b, then a + b is-

(A) 3 (B) 4

(C) 5 (D) 6

Q2. The equation of the curve which is such that the portion of the axis of x cut off between the origin and tangent at any point is proportional to the ordinate of that point is- (a & b are arbitrary constants)

(A) $x = y(a - b\log y)$

(B) $\log x = by^2 + a$

(C) $x^2 = y(a - b\log y)$

(D) None of these

Q3. The differential equation of all circles passing through the origin and having their centres on the x-axis is-

(A) $x^2 = y^2 + xy\frac{dy}{dx}$ (B) $x^2 = y^2 + 3xy\frac{dy}{dx}$

(C) $y^2 = x^2 + 2xy\frac{dy}{dx}$ (D) $y^2 = x^2 - 2xy\frac{dy}{dx}$

Q4. The differential equation of the family of circles with fixed radius 5 units and centre on the line y = 2 is-

(A) $(x - 2)y'^2 = 25 - (y - 2)^2$

(B) $(y - 2)y'^2 = 25 - (y - 2)^2$

(C) $(y - 2)^2y'^2 = 25 - (y - 2)^2$

(D) $(x - 2)^2y'^2 = 25 - (y - 2)^2$

Q5. A particular solution of $\log\frac{dy}{dx} = 3x + 4y$, $y(0) = 0$ is-

(A) $e^{3x} + 3e^{-4y} = 4$ (B) $4e^{3x} - e^{-4y} = 3$

(C) $3e^{3x} + 4e^{4y} = 7$ (D) $4e^{3x} + 3e^{-4y} = 7$

Q6. Number of values of $m \in N$ for which $y = e^{mx}$ is a solution of the differential equation $D^3y - 3D^2y - 4Dy + 12y = 0$ is -

(A) 0 (B) 1

(C) 2 (D) more than 2

Q7. A curve passes through the point (5, 3) and at any point (x, y) on it, the product of its slope and the ordinate is equal to its abscissa. The curve is a-

(A) Parabola (B) Ellipse

(C) Hyperbola (D) Circle

Q8. Let f(x) be differentiable on the interval $(0, \infty)$ such that f(1) = 1, and

$\lim_{t\to x}\frac{t^2f(x) - x^2f(t)}{t - x} = 1$, for each $x > 0$.

Then f(x) is -

(A) $\frac{1}{3x} + \frac{2x^2}{3}$ (B) $\frac{-1}{3x} + \frac{4x^2}{3}$

(C) $\frac{-1}{x} + \frac{2}{x^2}$ (D) $\frac{1}{x}$

Q9. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1 - y^2}}{y}$ determines family of circles with

(A) variable radii and a fixed centre at (0, 1)

(B) variable radii and a fixed centre at (0, –1)

(C) fixed radius 1 and variable centres along the x-axis

(D) fixed radius 1 and varialble centres along the y-axis

Q10. If $\left(\frac{2+\sin x}{1+y}\right)\frac{dy}{dx} = -\cos x, y(0) = 1,$

then $y\left(\frac{\pi}{2}\right) =$

(A) 1 (B) 1/2

(C) 1/3 (D) 1/4

Q11. If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is-

(c - arbitrary constant)

(A) $y\log\left(\frac{x}{y}\right) = cx$ (B) $x\log\left(\frac{y}{x}\right) = cy$

(C) $\log\left(\frac{y}{x}\right) = cx$ (D) $\log\left(\frac{x}{y}\right) = cy$

Q12. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is-

(A) $y = \ell nx + x$

(B) $y = x\ell nx + x^2$

(C) $y = xe^{(x-1)}$

(D) $y = x\ell nx + x$

Q13. The solution of primitive integral equation $(x^2 + y^2)\,dy = xy\,dx$, is $y = y(x)$.

If $y(1) = 1$ and $y(x_0) = e$, then x_0 is -

(A) $\sqrt{2(e^2-1)}$ (B) $\sqrt{2(e^2+1)}$

(C) $\sqrt{3}.e$ (D) none of these

Q14. For the primitive integral equation $ydx + y^2dy = x\,dy$; $x \in R, y > 0, y = y(x)$, $y(1) = 1$, then $y(-3)$ is-

(A) 3 (B) 2

(C) 1 (D) 5

Q15. A curve passes through the point $\left(1, \frac{\pi}{4}\right)$ & its slope at any point is given by

$\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$.

Then the curve has the equation -

(A) $y = x\tan^{-1}\left(\ell n\frac{e}{x}\right)$

(B) $y = x\tan^{-1}(\ell n + 2)$

(C) $y = \frac{1}{x}\tan^{-1}\left(\ell n\frac{e}{x}\right)$

(D) none

Q16. The solution of $(x - y^3)dx + 3xy^2dy = 0$ is-

(A) $\log x + \frac{x}{y^3} = k$ (B) $\log x + \frac{y^3}{x} = k$

(C) $\log x - \frac{x}{y^3} = k$ (D) $\log xy - y^3 = k$

Q17. The solution of $y^5 x + y - x\frac{dy}{dx} = 0$ is -

(A) $x^4/4 + 1/5\,(x/y)^5 = C$

(B) $x^5/5 + (1/4)\,(x/y)^4 = C$

(C) $(x/y)^5 + x^4/4 = C$

(D) $(xy)^4 + x^5/5 = C$

Q18. The solution of $\frac{dy}{dx} = \frac{ax+h}{by+k}$ represents a parabola when-

(A) $a = 0, b = 0$ (B) $a = 1, b = 2$

(C) $a = 0, b \neq 0$ (D) $a = 2, b = 1$

Q19. The solution of the differential equation, $e^x(x + 1)dx + (ye^y - xe^x)dy = 0$ with initial condition $f(0) = 0$, is -

(A) $xe^x + 2y^2e^y = 0$

(B) $2xe^x + y^2e^y = 0$

(C) $xe^x - 2y^2e^y = 0$

(D) $2xe^x - y^2e^y = 0$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The value of the constant 'm' and 'c' for which $y = mx + c$ is a solution of the differential equation $D^2y - 3Dy - 4y = -4x$.

(A) is $m = -1$

(B) is $c = 3/4$

(C) is $m = 1$

(D) is $c = -3/4$

Q2. The solution of the differential equation $(x + y)\,dy - (x - y)dx = 0$ is

(A) $y^2 + 2xy + x^2 = k$

(B) $y^2 + 2xy - x^2 = k$

(C) $-\left(-y^2 - 2xy + x^2\right) = k$

(D) $y^2 - 2xy + x^2 = k$

Q3. Solution of the equation $\frac{dy}{dx} + \frac{1}{x}\tan y = \frac{1}{x^2}$ $\tan y \sin y$ is

(A) $2x = \sin y(1 + 2cx^2)$

(B) $2x = \sin y(1 + cx^2)$

(C) $2x + \sin y(1 + cx^2) = 0$

(D) None of these

Q4. The differential equation

$\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \sin y + x^2 = 0$ is of the following type

(A) linear

(B) homogeneous

(C) order two

(D) degree one

Q5. Consider the differential equation, $y\,dx + (x + x^2 y)\,dy = 0$. , then

(A) Solution of the given equation is

$$-\frac{1}{xy} + \log y = C$$

(B) given equation is linear differential equation

(C) given equation is homogeneous differential equation

(D) given equation is neither homogeneous nor linear differential equation

Q6. Solution of the differential equation $\left(1 + y^2\right)dx = \left(\tan^{-1} y - x\right)dy$ is

(A) $xe^{\tan^{-1} y} = \left(1 - \tan^{-1} y\right)e^{\tan^{-1} y} + c$

(B) $xe^{\tan^{-1} y} = \left(\tan^{-1} y - 1\right)e^{\tan^{-1} y} + c$

(C) $x = \tan^{-1} y - 1 + ce^{-\tan^{-1} y}$

(D) None of the above

(where c is arbitrary constant)

Q7. The function $f(x)$ satisfying the equation, $f^2(x) + 4f'(x) \cdot f(x) + [f'(x)]^2 = 0$ is -

(A) $f(x) = c \cdot e^{(2-\sqrt{3})x}$

(B) $f(x) = c \cdot e^{(2+\sqrt{3})x}$

(C) $f(x) = c \cdot e^{(\sqrt{3}-2)x}$

(D) $f(x) = c \cdot e^{-(2+\sqrt{3})x}$

Q8. Let C be a curve such that the tangent at any point P on it meets x-axis and y-axis at A and B respectively. If BP : PA = 3:1 and the curve passes through the point (1,1), then

(A) The curve passes through $\left(2, \frac{1}{8}\right)$

(B) Equation of normal to the curve at (1,1) is $3y - x = 2$

(C) The differential equation for the curve is $3y' + xy = 0$

(D) The differential equation for the curve is $xy' + 3y = 0$

Q9. The tangent at any point P on a curve $f(x, y) = 0$ cuts the y-axis at T. If the distance of the point T from P equals the distance of T from the origin then the curve with this property represents a family of circles. Which of the following is/are **CORRECT**?

(A) Any arbitrary line $y = mx$ cuts every member of this family at the points where the slopes of these members are equal.

(B) $f(x, y) = 0$ is orthogonal to the family of circles $x^2 + y^2 - ky = 0 \ \forall \ k \in R$.

(C) If $f(x, y) = 0$ passes through (2, 2) then the intercept made by its director circle on the y-axis is equal to 8.

(D) If $f(x, y) = 0$ passes through $(-1, 1)$ then image of its centre in the line $y = x$, is $(1, 0)$.

Q10. Let C be the family of curves $f(x, y, c) = 0$ (no member of C is x-axis) such that length of subnormal at any point P(x, y) on the curve C is equal to four times that of the length of subtangent at the same point.

Which of the following statement(s) is(are) correct ?

(A) Equation of the line with positive y-intercept passing through (4, 2) and perpendicular to the curve C is $x + 2y = 8$.

(B) Orthogonal trajectory of C is family of parallel lines having gradient ± 2.

(C) Order and degree of the differential equation of family of curves C are 1 and 2 respectively.

(D) Differential equation of family of curves is $2y' \pm x = 0$.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 11 to 12)

Newton's law of cooling states that the rate at which a substance cools in moving air is proportional to the difference between the temperature of the substance and that of the air. If the temperature of the air is 290 K. We can write it as $\frac{dT}{dt} = -k(T - 290)$, $k > 0$ constants. where T is temperature of substance.

The substances cools from 370 K to 330 K in 10 min. On the basis of above information, answer the following question.

Q11. Solution of differential equation is -

(A) $T = 290 + 160\ e^{-kt}$

(B) $T = 290 + 80e^{-kt}$

(C) $T = 290 + 40\ e^{-kt}$

(D) $T = 290 + 20\ e^{-kt}$

Q12. The value of k must be

(A) $\ell n\ 2$ (B) $\ell n2/40$

(C) $\ell n2/20$ (D) $\ell n2/10$

Paragraph # 2 (Ques. 13 to 14)

Let f be a real valued differentiable function satisfying $f(x + y) = f(x) + f(y) - xy - 1, \ \forall \ x, y \in R$ and $f(1) = 1$.

Q13. Equation of the normal to the graph of $f(x)$ at the point where $y = f(x)$ cuts the y-axis, is

(A) $2x - y + 1 = 0$ (B) $2x + y - 1 = 0$

(C) $x - 2y + 2 = 0$ (D) $x + 2y - 2 = 0$

Q14. Area enclosed by the curve $y = f(x)$ in the first quadrant is

(A) $\frac{5}{3}$ (B) $\frac{4}{3}$

(C) $\frac{5}{4}$ (D) $\frac{9}{4}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. Match the properties of the curves given in column-I with the corresponding curve(x) given in the column-II.

Column-I	**Column-II**
(A) A curve passing through (2, 3) having the property that length of the radius vector of any of its point P is equal to the length of the tangent drawn at this point, can be	(P) Straight line
(B) A curve passing through (1, 1) having the property that any tangent intersects the y-axis at the point which is equidistant from the point of tangency and the origin, can be	(Q) Circle
(C) A curve passing through (1, 0) for which the length of normal is equal to the radius vector, can be	(R) Parabola
(D) A curve passes through the point (2, 1) and having the property that the segment of any of its tangent between the point of tangency and the x-axis is bisected by the y-axis, can be	(S) Hyperbola

Q16. Match the following

Column I	**Column II**
(A) Solution of $y-\frac{xdy}{dx}=y^2+\frac{dy}{dx}$, is	(P) $xy^2=2y^5+c$
(B) Solution of $(2x-10y^3)\frac{dy}{dx}+y=0$, is	(Q) $\sec y = x+1+ce^x$
(C) Solution of $\sec^2 y\,dy + \tan y\,dx = dx$, is	(R) $(x+1)(1-y)=cy$
(D) Solution of $\sin y\frac{dy}{dx}=\cos y\,(1-x\ \cos y)$, is	(S) $\tan y = 1+ce^{-x}$

Q17. Let a function $y = f(x)$ satisfies the following conditions

$\frac{dy}{dx}=y+\int_0^1 ydx$ and $f(0)=1$

Column-I	**Column-II**
(A) $f'(0)$ is equal to	(P) $f(0)$
(B) $f(1)$ is equal to	(Q) $\frac{2}{3-e}$
(C) $\lim_{x\to 0}\frac{f(x)-1}{x}$ is equal to	(R) $\frac{e+1}{3-e}$
(D) $\frac{1}{2}f'(\ell n(3-e))$ is equal to	(S) $f'(0)$
	(T) 1

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column I (family of curve)	Column II (differential equation)	Column III (sum of order and degree)
(A) $y = kx + k^2$	(P) $1 - \left(\frac{dy}{dx}\right)^2 = \frac{2x}{y}\cdot\frac{dy}{dx}$	(L) 1
(B) $y = ae^{2x} + be^{3x}$	(Q) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 = 2$	(M) 2
(C) $y^2 = 4a(x + a)$	(R) $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$	(N) 3
(D) $xy = ae^x + be^{-x} + x^2$	(S) $y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$	(O) Not defined

Q18. Which of the following is only correct combination?

(A) A-P-N (B) A-Q-N
(C) A-R-N (D) A-S-N

Q19. Which of the following is only correct combination?

(A) B-P-N (B) B-Q-N
(C) B-R-N (D) B-Q-M

Q20. Which of the following is only correct combination?

(A) C-P-N (B) C-Q-N
(C) C-R-N (D) C-R-M

Q21. Which of the following is only correct combination?

(A) D-R-L (B) D-Q-N
(C) D-P-M (D) D-P-O

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q22. Let $y = f(x)$ be a curve C_1 passing through $(2,2)$ and $\left(8,\frac{1}{2}\right)$ and satisfying a differential equation $y\left(\frac{d^2y}{dx^2}\right) = 2\left(\frac{dy}{dx}\right)^2$. Curve C_2 is the director circle of the circle $x^2 + y^2 = 2$. If the shortest distance between the curves C_1 and C_2 is $\left(\sqrt{p} - q\right)$ where $p, q \in N$, then find the value of $(p^2 - q)$.

Q23. If $x = \int_{-y}^{y} \frac{dt}{\sqrt{1+9t^2}}$ and $\frac{d^2y}{dx^2} = ky$, then 4k equals

SUBJECTIVE TYPE

Q24. A curve passes through (2,0) and the slope of tangent at point P(x,y) equals $\frac{(x+1)^2 + y - 3}{(x+1)}$.

Find the equation of the curve and area enclosed by the curve and the x-axis in the fourth quadrant.

Q25. The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being $k > 0$. Find

(a) Population at any time t, given initial population of the town being 2500.

(b) If 10 years later the population has fallen to 1900, find the time when the population will be 1500.

(c) Predict about the population of the town in the long run.

Q26. It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass of the radius was m_0 and during time t_0 α % of the original mass of radium decay.

Q27. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min, and the mixture is pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

Q28. A curve passing through (1,0) such that the ratio of the square of the intercept cut by any tangent off the y-axis to the subnormal is equal to the ratio of the product of the co-ordinates of the point of tangency to the product of square of the slope of the tangent and the subtangent at the same point. Determine all such possible curves.

Q29. A tank consists of 50 litres of fresh water. Two litres of brine each litre containing 5 gms of dissolved salt are run into tank per minute; the mixture is kept uniform by stirring, and runs out at the rate of one litre per minute. If 'm' grams of salt are present in the tank after t minute, express 'm' in terms of t and find the amount of salt present after 10 minutes.

Q30. Let C be a curve passing through M (2, 2) such that the slope of the tangent at any point to the curve is reciprocal of the ordinate of the point. If the area bounded by curve C and line $x = 2$ is expressed as a rational $\frac{p}{q}$ (where p and q are in their lowest form), then find $(p + q)$.

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
A	A	D	C	B	C	A	B	B	A
11	12	13	14	15	16	17	18	19	20
C	C	A	A	B	A	C	A	A	A
21	22								
A	B								

Ex - 1A

1. A,C,D **2.** A,D **3.** A,B,C **4.** A,B **5.** A,B

6. B **7.** B **8.** D **9.** A **10.** C

11. C **12.** C **13.** D **14.** C

15. A → Q; B → S; C → P; D → S **16.** C **17.** A

18. B **19.** A **20.** $x\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]=y\frac{dy}{dx}$

21. $(x^2 - y^2)\frac{dy}{dx} - 2xy = 0$ **22.** $y = -\frac{6}{x} + c.$

23. $x\frac{dy}{dx} + a\frac{dx}{dy} = x(c) + a\frac{1}{c} = y$ **24.** $y = -\text{cosec}x + \cot x + c$

25. $y = x\tan(c - x)$ **26.** $\log\left(\frac{x}{y}\right) = \frac{1}{2}\left(\frac{1}{y^2} + \frac{1}{x^2}\right) + c$

27. $-\frac{1}{e^y} = e^x + \frac{1}{3}x^3 + c$ **28.** $\frac{e^{2x}}{2} + \frac{1}{y} + x + c = 0$

29. $\tan y - \cot x = c$ **30.** $\tan\left(\frac{x+y}{2}\right) = x + C$

31. $\log_e(x + y) = x - y + c$ **32.** $y\sec x = \tan x + c$

33. $y\frac{1}{(1-x^2)} = \frac{1}{\sqrt{1-x^2}} + c$ **34.** $y^{-5}x^{-5} = \frac{5}{2}x^{-2} + c$

35. $y = \log\sec x \pm c_1 x \pm c_2$

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	A	C	C	D	C	C	A	C	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	
C	D	C	A	A	B	B	C	B	

EX - 2A

1. C, D **2.** B, C **3.** A, B **4.** C, D **5.** A,D
6. B, C **7.** C, D **8.** A,B,D **9.** A,B **10.** A,C
11. B **12.** D **13.** B **14.** A
15. A → P,S; B → Q; C → Q, S; D → R
16. A → R; B → P; C → S; D → Q
17. A → Q,S; B → R; C → Q, S; D → P,T **18.** D **19.** C
20. A **21.** B **22.** 0062 **23.** 9 **24.** $\frac{4}{3}$

25. (a) $P = 1000 + 1500\,e^{-kt}$ where $k = \frac{1}{10}\ln\left(\frac{5}{3}\right)$
(b) $T = 10\log_{5/3}(3)$ (c) $P = 1000$ as $t \to \infty$]

26. $m = m_0 e^{-kt}$ where $k = \frac{-1}{t_0}\ln\left(1 - \frac{\alpha}{100}\right)$

27. $27\frac{7}{9}$ minutes **28.** $x = e^{2\sqrt{y/x}}$; $x = e^{-2\sqrt{y/x}}$ **29.** $y = 5t\left(1 + \frac{50}{50+t}\right)$gms; $91\frac{2}{3}$ gms

30. 19

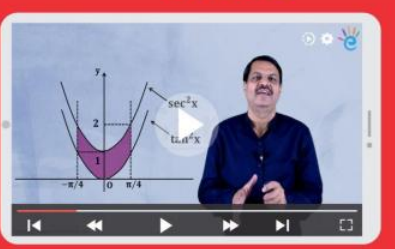
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